

Selectime: expanded input-selection study

19 September 2026

Main findings

On 90 matched TIME tasks, the best aggregate result is the frozen **Chronos-2 scope mixture**: mean task MASE 1.058165, or **1.03% lower** than univariate Chronos-2. It is only 0.08% below native multivariate Chronos-2, so the practical difference between those two means is small. Per-variate and task selection improve on univariate by 0.83% and 0.49%, respectively. The standalone retrieved-covariate candidates are all worse than univariate, and increasing K from 1 to 20 adds substantial recorded inference cost without an aggregate accuracy gain.

Chronos-Bolt univariate has mean task MASE 1.167905, 9.23% above univariate Chronos-2. The direct-horizon mixture is not usable as an aggregate method: one SG_Carpark task reaches MASE 313,637 with Chronos-2 and 382,465 with Bolt, dominating both 90-task means. The remaining findings are not driven by this method.

1 Question and protocol

For an observed target history x and horizon- H future y , a frozen backbone maps its supplied input representation to a median point forecast $\hat{y}$. Selectime asks whether a held-out validation period can choose or mix useful inputs without training model parameters and without observing the official test future.

Chronos-2 compares univariate, native multivariate, past-only $\sqrt{|x|}$ /sign augmentation, and retrieved covariates for $K \in \{1, 5, 10, 15, 20\}$. It also reports task and per-variate candidate selection, binary scope selection, a frozen univariate/multivariate mixture, a frozen univariate/top-5 mixture, and a direct-neighbor-horizon mixture. Chronos-Bolt reports univariate forecasts and the direct-horizon mixture.

Validation choices use mean squared scaled error and are frozen before test. Validation rows with no finite available target history are excluded before backbone inference; if no usable rows remain, selection keeps univariate input. Soft mixtures use a Beta(1,1) smoothed validation win frequency with half ties and exclude fallback rows. Retrieval uses instance-normalized lookback distance, aligned historical windows, and lookback-only rescaling. Candidate fallbacks use the canonical univariate forecast. The comparison uses seed 0 once. Each method has 67,502 finite MASE cells on the same Seasonal-defined grid; the reported mean gives equal weight to each of 90 task means.

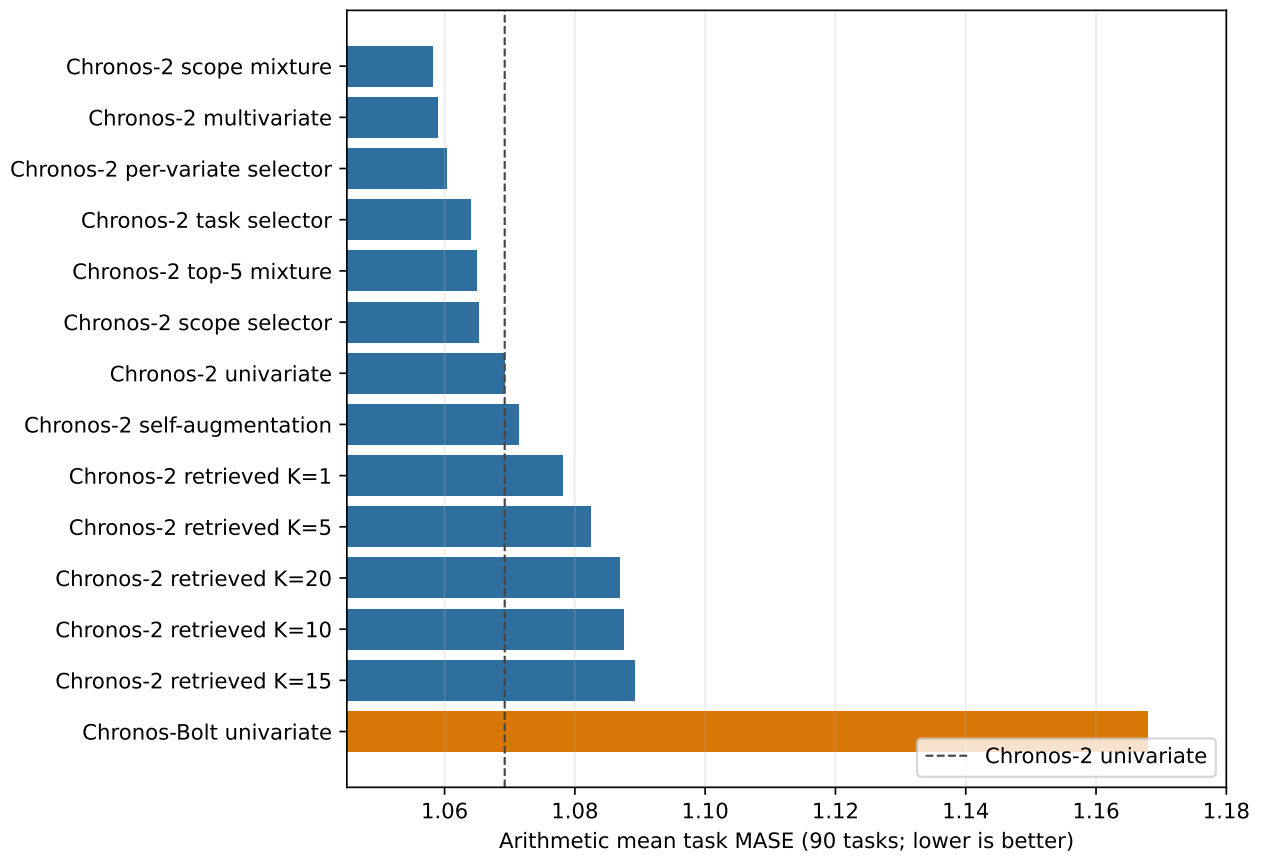


Figure 1: Raw arithmetic mean task MASE for every stable reported method. Direct-horizon mixtures are separated because their outlier would make the other bars unreadable. Lower is better.

2 Obtained accuracy

Method	Mean MASE	Scaled MASE	Δ vs C2 uni.	Win/loss/tie
C2 scope mixture	1.058165	0.701774	−1.03%	56/10/24
C2 multivariate	1.058981	0.703148	−0.96%	43/23/24
C2 per-variate selector	1.060312	0.703046	−0.83%	55/27/8
C2 task selector	1.063977	0.704353	−0.49%	35/16/39
C2 top-5 mixture	1.064982	0.705235	−0.40%	49/27/14
C2 scope selector	1.065183	0.705885	−0.38%	24/10/56
C2 univariate	1.069230	0.708652	0.00%	–
C2 self-augmentation	1.071291	0.711580	+0.19%	42/48/0
C2 retrieved $K = 1$	1.078047	0.712306	+0.82%	30/52/8
C2 retrieved $K = 5$	1.082413	0.715548	+1.23%	27/55/8
C2 retrieved $K = 20$	1.086914	0.716907	+1.65%	32/50/8
C2 retrieved $K = 10$	1.087506	0.717717	+1.71%	29/53/8
C2 retrieved $K = 15$	1.089185	0.717431	+1.87%	30/52/8
Bolt univariate	1.167905	0.776717	+9.23%	8/82/0

“Scaled MASE” divides each task MASE by its matched held-out Seasonal Naive MASE before averaging. Win/loss/tie is paired against univariate Chronos-2 on the same tasks. Scope mixing wins on 56 tasks and loses on 10, but the small aggregate margin over native multivariate input does not establish a meaningful or statistically significant superiority.

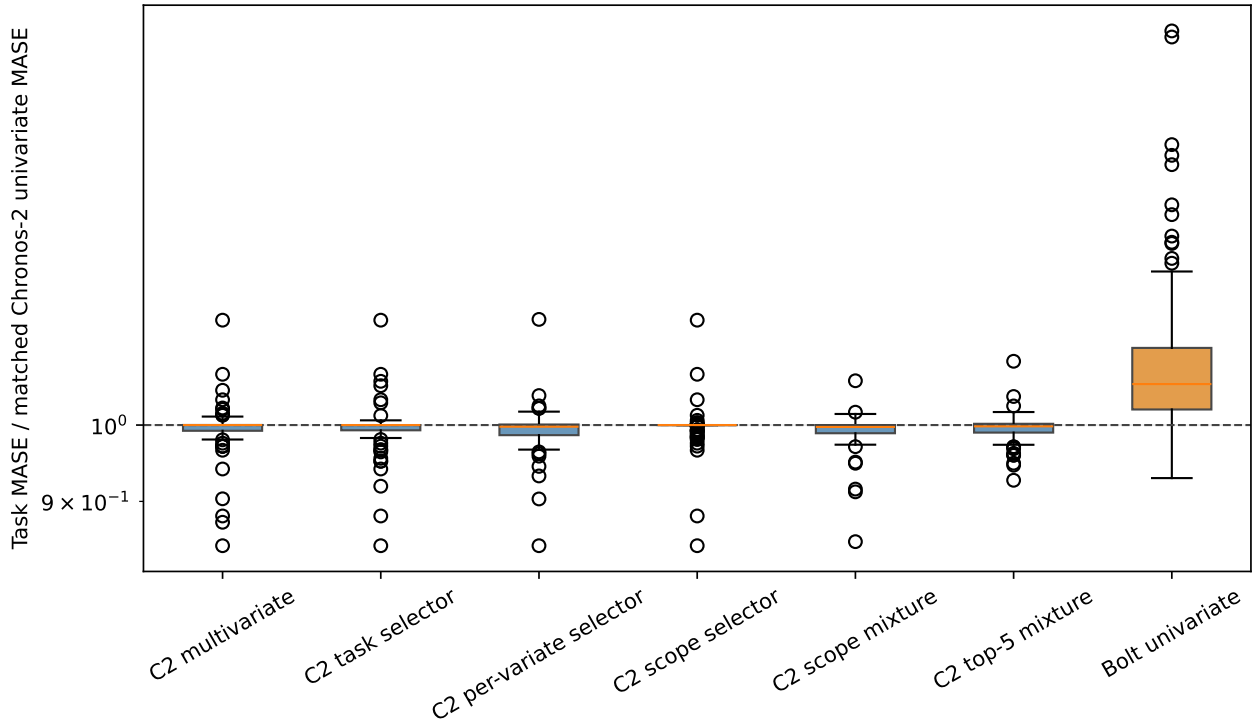


Figure 2: Paired task MASE ratios. The dashed line is parity with univariate Chronos-2; the vertical scale is logarithmic. The boxes show heterogeneous task effects hidden by the aggregate mean.

3 What validation selected

The task selector keeps univariate input on 39 of 90 tasks, chooses multivariate on 31, self-augmentation on nine, and retrieval on eleven. Its retrieval choices include four $K = 20$ tasks. The 3,882 per-variate

decisions use univariate input 51.4% of the time and multivariate input 22.3%; all retrieval choices together account for 14.9%.

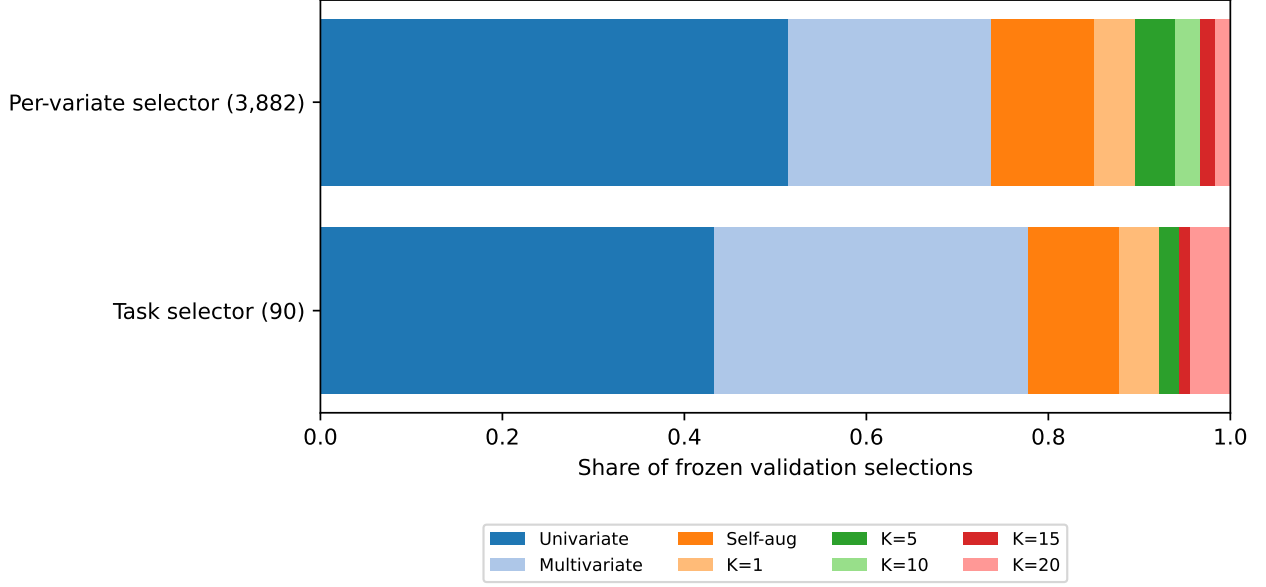


Figure 3: Frozen validation choices for the original task and per-variate selectors. These frequencies describe selected representations, not causal contributions to accuracy.

Every standalone retrieved- K method uses univariate fallback on 10,293 cells (15.25%). The task and per-variate selectors fall back on 6 and 440 cells; the top-5 mixture falls back on 3,032 cells. Complete metric coverage therefore does not imply successful retrieval everywhere.

4 Accuracy and recorded cost

The summed Chronos-2 candidate query-loop times are 235 seconds for univariate, 247 for multivariate, 736 for self-augmentation, and 433, 1,406, 2,710, 3,950 and 5,279 seconds for $K = 1, 5, 10, 15, 20$. These are not end-to-end times: retrieval and datastore preprocessing are recorded separately, and selected methods reuse candidate artifacts rather than having an independently measured latency.

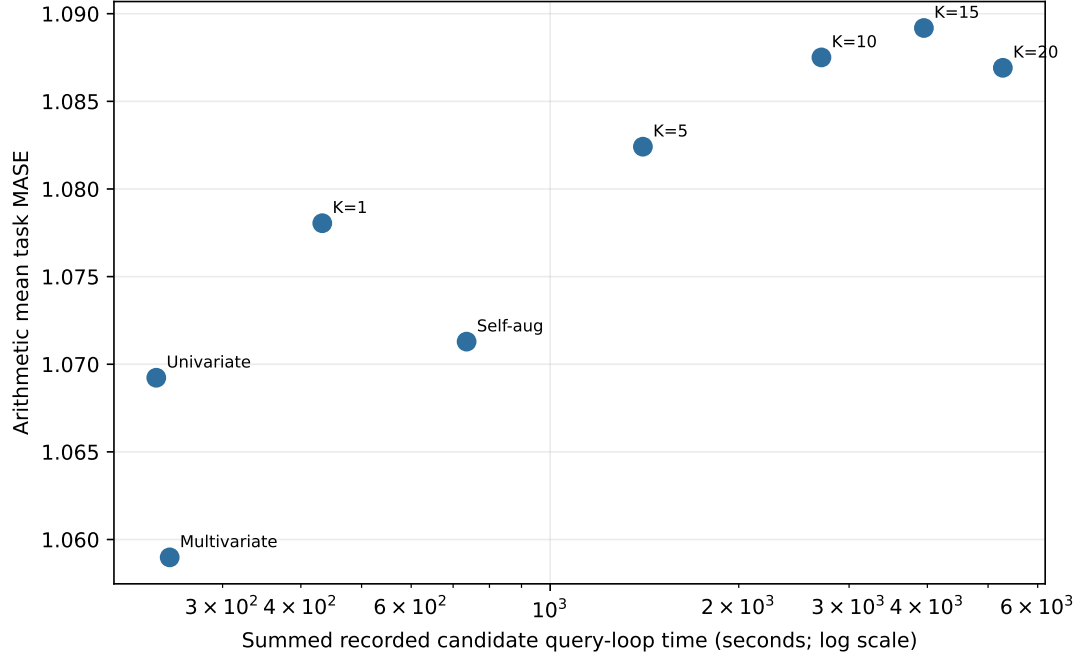


Figure 4: Candidate accuracy versus recorded query-loop time. Larger retrieved sets are slower and do not improve the 90-task mean.

5 Direct-horizon instability

The direct-horizon method rescales a retrieved observed continuation using neighbor and query lookback standard deviations, then mixes it with the univariate forecast. Its Chronos-2/Bolt aggregate MASE is 3,485.98/4,250.80. Excluding only SG_Carpark/15T/short reduces those means to 1.1471/1.2139, showing that one extreme task dominates, although even the reduced means do not beat the corresponding vanilla forecasts.

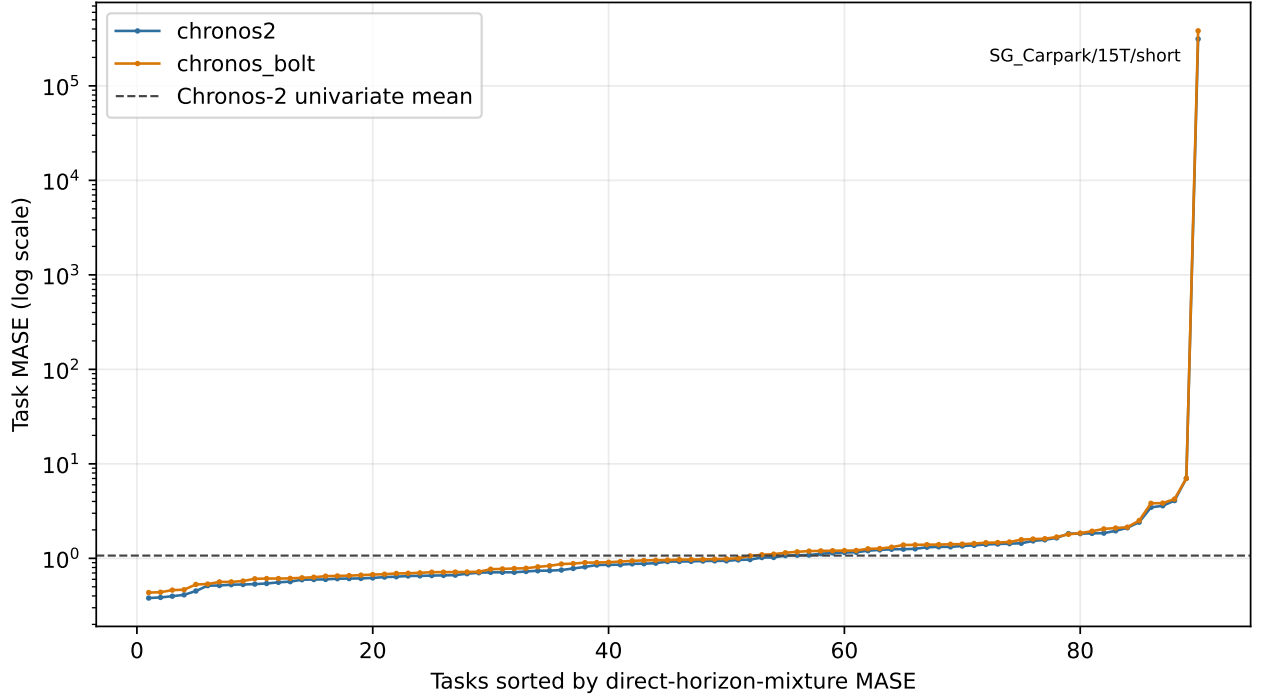


Figure 5: Sorted direct-horizon-mixture task MASE. The log scale exposes the single six-order-of-magnitude failure while retaining the other tasks.

The compact synchronized evidence proves the extreme finite errors and the unbounded rescaling rule, but it does not include raw neighbor or prediction arrays. The exact offending neighbor/channel cannot therefore be reconstructed locally. The supported conclusion is that this direct-horizon formulation is numerically unsafe as evaluated; attributing the outlier to a specific near-constant lookback remains a plausible mechanism, not an observed fact.

6 Limits

The report manifests and compact task summaries agree with the expanded scientific contract. Raw prediction, retrieval and metric arrays were not independently inspected. One seed provides no repeat robustness or significance assessment. Population standard deviations describe within-task metric cells, not uncertainty across runs. Selected-method latency remains undefined. These limits do not change the observed ordering, but they constrain claims about generalization, uncertainty and execution cost.